



# Effect of Flexible Inverted PVDF in Free Shear Layer Wake

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The use of flexible inverted piezo embedded Polyvinylidene Difluoride (PVDF) as a simultaneous energy harvester and as a sensor to detect changes in the wake is explored. The oscillation amplitude (characterized by voltage output) and oscillation frequency of the piezo embedded PDVF was quantified in the wake of NACA 0012 wall-to-wall model and SD7003 wall-to-wall model at a Reynolds number of 100,000 and 67,000 respectively. The performance of the sensor was also quantified in the freestream without the presence of the wing. In order to quantify the sensor response to angle of attack and downstream distance, the amplitude and frequency of oscillations was recorded in the wing wake. Increase in angle of attack of the wing resulted in increase in oscillation frequency and amplitude of the PVDF. It also indicated that the inverted flag configuration performed better in the wake under unsteady conditions when compared to freestream conditions. Using Particle Image Velocimetry (PIV), it was established that the wake signature was not affected by the presence of PVDF which enticed the possibility of using inverted PVDF as a sensor to detect changes in wake signature. The RMS contours of voltage in the wake of SD7003 airfoil show remarkable wake features such as upper and lower surface stratification and downwash angle which shows the sensitivity of the sensor to the unsteadiness in the wake. The capability of this device to act as a potential energy harvester and as a sensor has serious implications in extending the mission capabilities of small UAVs.

## I. Nomenclature

|            |   |                                       |
|------------|---|---------------------------------------|
| $c$        | - | Chord length (m)                      |
| $x, y, z$  | - | Coordinate systems (m)                |
| $\rho$     | - | Density (Kg/m <sup>3</sup> )          |
| $U_\infty$ | - | Freestream velocity (m/s)             |
| $L$        | - | Membrane length (m)                   |
| $t$        | - | Membrane thickness (m)                |
| $\beta$    | - | Non-dimensional bending stiffness     |
| $A$        | - | Oscillation Amplitude (m)             |
| $f$        | - | Oscillation frequency (Hz)            |
| $\nu$      | - | Poisson's ratio                       |
| $u'$       | - | Streamwise Fluctuating Velocity (m/s) |
| $U_x$      | - | Streamwise Velocity (m/s)             |
| $v'$       | - | Transverse Fluctuating Velocity (m/s) |
| $V$        | - | Voltage (V)                           |
| $L_0$      | - | Wake Half Width (m)                   |
| $E$        | - | Young's Modulus (Pa)                  |
| $PVDF$     | - | Polyvinylidene Difluoride             |
| $RMS$      | - | Root Mean Square                      |

## II. Introduction

The usage of drones (small scaled unmanned aerial vehicles) have increased exponentially in the recent years especially in personal, commercial, entertainment, industrial and military applications. The higher usage of drones is predicated on their capability to reach remote areas and require least amount of time, effort and energy. However, some of the major hindrances in the use of drones is its lower battery life, less maneuverability and poor

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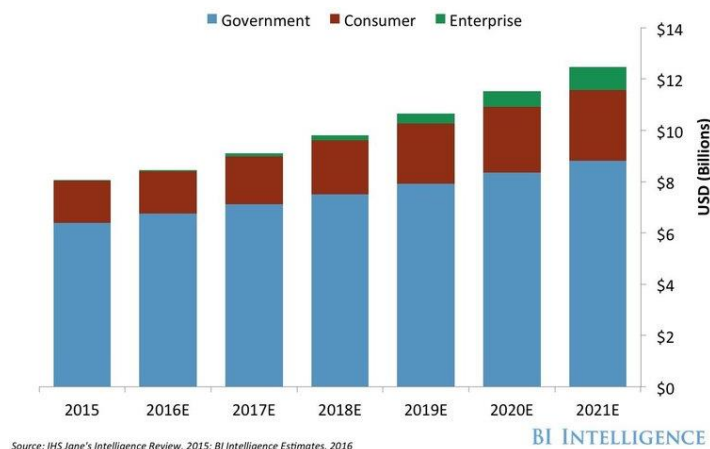
gusts response. Due to restrictions in size and weight of the drones, increasing the flight time and stability remains a velleity. Any incremental gain in battery life will translate to longer flight time. And any alternative ways to achieve control rather than using traditional control surfaces will save weight. The current research explores the possibility of overcoming these shortcomings by employing flexible inverted piezo embedded Polyvinylidene Difluoride (PVDF) in the wake of a wing to 1) Harvest energy to increase battery life. 2) Sense changes in the wing wake for gust alleviation and 3) Increase maneuverability using PVDFs as distributed control surfaces.

Globally, the use of small-scale drones has been increasing in an exponential rate. In 2018, the estimated investments in drones' totals over 10 billion US dollars (as shown in Figure 1) and will continue to grow in the near future [1]. As such, increasing the reliability, endurance, range and flight time of the drones is given significant priority by government and non-government organizations. Therefore, it is prudent that continuous effort is put into developing technologies to overcome the shortcomings of the drones. Some of the shortcomings are mentioned below:

**A) Lower Battery Life:** Most drones have a charge time of 50-120 minutes but can only stay in the air only for 15-30 min [2]. The flight time reduces even further if there are other equipment such as cameras, GPS, etc. Therefore, any incremental gain in battery life will translate to longer flight time.

**B) Gust Response:** Due to the small scale and low altitude flight, drones are highly susceptible to gust created by buildings, atmospheric boundary layer, thermal inversions and clear air turbulence (CAT). Without electronics, the drones are inherently unstable due to its high center of gravity, low moment of inertia and high wind sensitivity [3]. Therefore, technology needs to be developed to sense the changes in the drone's flight path, changes in wind speed and wing performance.

**C) Maneuverability:** Depending on the type of drone, (fixed wing or rotary) the maneuverability of the aircraft varies. Fixed wing drones use conventional control surfaces (ailerons, rudders and elevators) to maneuver whereas rotary drones can maneuver by changing the thrust from each rotor. As such, rotary drones are more agile than fixed wing, but fixed wing drones can fly the farthest since most of the lifting work is done by the wings and not by the rotors. Therefore, a non-conventional distributed control system is needed to increase the maneuverability of the fixed wing drones.



**Figure 1 Estimated investment in drones is USD. As the need and application for drones increase over the years, the investment also increases. [1]**

This research is an endeavor to address the listed shortcomings of the drones by exploring the use of piezo embedded Polyvinylidene Difluoride (PVDF) in the wing wake. Since PVDFs are smart materials, they produce capacitance when deformed but can also be intentionally actuated by applying voltage difference. The latter capability of the PVDF allows it to act as a potential control surface. However, the possibility of using PVDF for maneuvering was not considered in this paper but will be presented in a follow up paper. The results presented in this paper focus on the energy harvesting and gust response only.

## A. Wing Wake

When flow from the upper surface meets the flow from the lower surface at the trailing edge, the resultant shear causes the air to roll up creating vortices which shed at a given frequency. The vortex shedding frequency in the wake is known to vary with the Reynolds number (Figure 2) [4]. Therefore, with different flight speed and conditions,

the wake signature changes for the same airfoil. Figure 2a and 2b shows prominent turbulent wake vortex shedding due to the separated upper surface shear layer. Huang & Lin [5] and Huang & Lee [6] performed experiments on NACA 0012 airfoil and reported that the vortex shedding is only observed at lower Reynolds numbers where boundary layer separation occurs without reattachment. Yarusevych et al. [4] amended this result and proved through hot-wire measurements that vortex shedding occurs even after boundary layer attaches to the surface at higher Reynolds number as shown in Figure 2c and that the vortex shedding varies *linearly* with the Reynolds number.

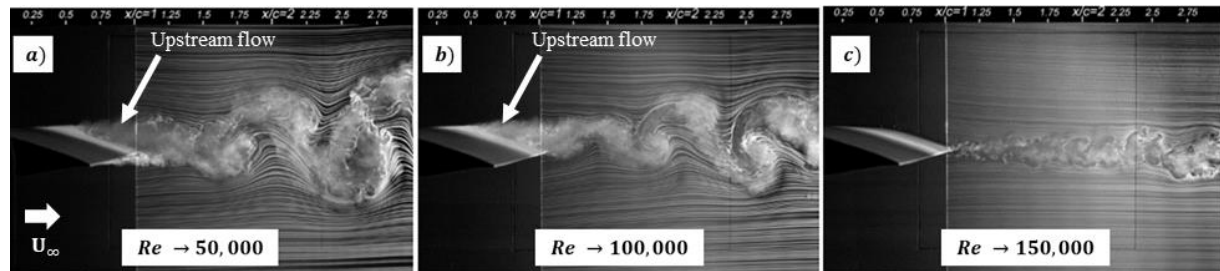


Figure 2 Shedding of vortices from the trailing edge of NACA 0025 airfoil at different Reynolds numbers [4]

The vortex shedding is also a function of trailing edge geometry. Guan et al. [7] experimented with multiple beveled trailing edge geometries and showed that even subtle changes in geometry can result in substantial changes in wake signatures. The vortex shedding is greater at the sharp trailing edge when compared to the smooth trailing edges. Even the smooth trailing edge was seen to have certain vortex shedding frequency found through two-point correlation of streamwise velocity component which complements the result from Yarusevych et al. [4]. Since the vortex shedding from wing is a function of multitude of variables such as angle of attack, Reynolds number, trailing edge geometry, airfoils, etc., any changes in the wing performance and wing aerodynamics will present itself as changes in the wing wake and therefore can be detected! It is the unique and sensitive nature of the wake behind the wing which allows the potential for sensing and energy harvesting.

## B. Flexible Smart Materials

Historically, smart materials have been used to harvest energy in variety of applications. Allen and Smits [8] studied the feasibility of placing a piezoelectric membrane or “eel” in the wake of a flat plate at  $90^\circ$  angle of attack and using the von Karman vortex street forming behind the bluff body to induce oscillations in the membrane (Figure 3). The oscillations result in a capacitive build up in the membrane that provides a voltage source that can be used for various purposes. The membrane is shown to have oscillated in the same frequency as the undisturbed vortex shedding in the wake.

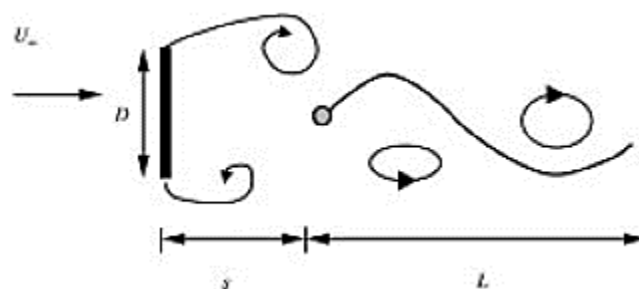
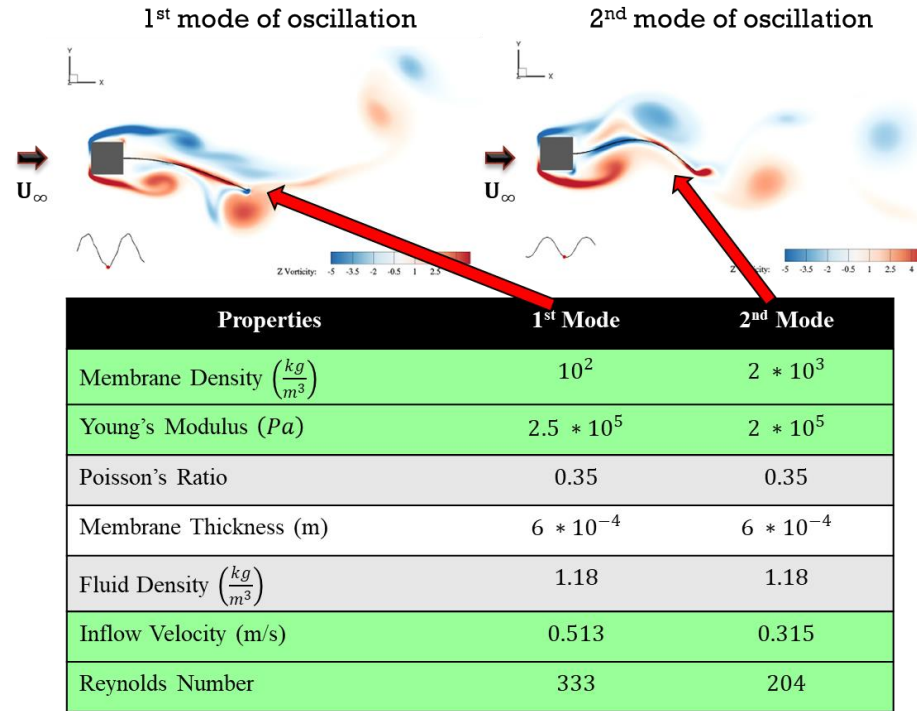


Figure 3 Geometry of oscillating membrane behind flat plate [8]

Similar relationship between the vortex shedding frequency and the frequency of oscillations were observed in Gilmanov et al. [9] But this dependency is a function of the Reynolds number based on the inflow velocity  $U_\infty$ , cylinder size, the geometric (cantilever length  $L$  and thickness  $t$ ), material (structure density) and structural (Young's modulus  $E$ , Poisson's ratio  $\nu$ ) properties. Based on these properties, the cantilevered membrane may develop complex vibration modes. Gilmanov et al. [9] did a LES study on the fluid – structure interaction of a thin membrane in the wake of a square cylinder were able to model the first mode of oscillations (closer to the natural frequency of the membrane) and the second mode of oscillation (kinematics resembling a traveling wave with a stagnation point). The differences in the input properties that contributed to the different modes of oscillations is highlighted in Figure 4. The 2<sup>nd</sup> oscillation mode was achieved by increasing the density of the membrane by a factor of 10 with minor changes in

Young's modulus,  $U_\infty$  and Reynolds number. Therefore, it is possible to tune the membrane to obtain oscillating conditions which are effective for energy production.



**Figure 4 Comparison of the 1<sup>st</sup> and 2<sup>nd</sup> oscillation modes of the membrane in the wake of a square cylinder. Multiple parameters are responsible for changes in different modes of oscillations but the dominant being the material density. [9]**

Similar to the studies above, there are a number of literature where a flexible material is used to determine the interaction between the vortex shedding and the oscillations in the wake of a bluff body in order to harvest energy. The massive vortex shedding behind bluff bodies provides large amplitude oscillations which results in higher capacitance buildup in the material. Streamline bodies such as wings with lower thickness do not experience massive vortex shedding as bluff bodies at lower angles of attack. Therefore, more often than not, flexible material or smart materials are used to reduce drag (Liou et al. [10] were able to reduce drag on a cylinder by 30% with a membrane length to chord ( $L/c$ ) ratio of 1.2, Stanewsky [11]), enhance lift (Neuhart and Pendergraft [12], Jang et al. [13], Storms and Jang [14], Traub [15]) and suppress flutter (Lee and Kroo [16]) in a streamlined body and not particularly for energy harvesting purposes. Because of the lower vortex shedding magnitude, flexible extended trailing edges like the one shown in Figure 4, might not be effective in harvesting energy from the wake of a streamlined body. Therefore, the use of inverted flag configurations (free leading edge and fixed trailing edge) is explored as a potential option to harvest energy from the wake of a streamlined body.

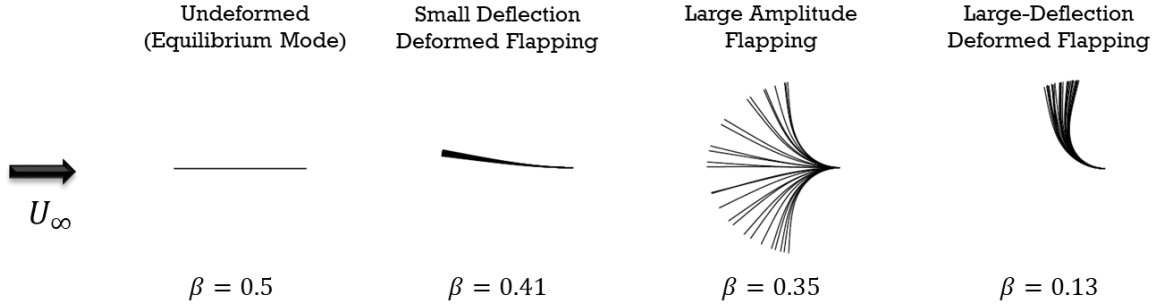
Recently, the inverted flag configuration has captured a lot of attention as an energy harvesting device. Depicted in Figure 5 (Goza and Colonius [17]), the oscillations of the inverted flag are a direct function of flow instabilities (destabilizing parameter) and the non-dimensional structural stiffness ( $\beta$ ) (stabilizing parameter) (Kim et al. [18]). The non-dimensional bending stiffness is determined by,

$$\beta = \frac{B}{\rho U_\infty^2 L^3} \quad (1)$$

where  $\rho$  is the density of the fluid,  $U_\infty$  is the freestream velocity,  $L$  is the length of the flag and  $B$  is the dimensional flexural rigidity of the flag which can be determined by

$$B = \frac{Et^3}{12(1-\nu^2)} \quad (2)$$

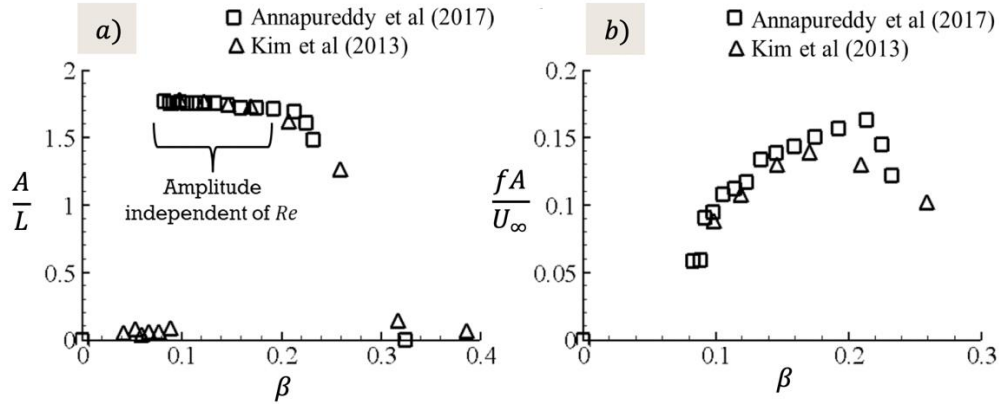
where  $E$  is the modulus of elasticity,  $t$  is the thickness of the membrane and  $\nu$  is the Poisson's ratio.



**Figure 5** A time lapse of the flag position of an inverted flag for different non-dimensional stiffness ( $\beta$ ) at a Reynolds number of 200. (Goza and Colonius [17]).

The inverted flag configurations have been tested over a range of Reynolds numbers from 200 to 100,000 using experimental and computational techniques. While some features of conventional flag flapping were found to be independent of the Reynolds number and three dimensional effects (Akcabay & Young, [19]), experiments done by Kim et al. [18] at  $Re = O(10^4)$ , Gurugubelli and Jaiman [20] and Goza and Colonius [17] at  $Re = O(100 - 1000)$  and computations done by Annapureddy et al. [21] at  $Re = 90,000$  show similar dynamical regimes which indicates some independence of inverted flag flapping on the Reynolds number.

This independence can be seen in the non-dimensional amplitude ( $A/L$ ) (Figure 6a) where the amplitude in the range of 1.7 – 1.8 is insensitive to Reynolds number. It is not surprising that the frequency (Figure 6b) changes with the Reynolds number as the inverted flag changes from equilibrium mode ( $\beta = 0.5$ ) to flapping mode ( $\beta < 0.4$ ) with increase in Reynolds number. However, the dependence of inverted flag oscillations in three-dimensional flow is an open question which warrants further investigation (Goza and Colonius [17]).



**Figure 6** Variation of non-dimensional flapping amplitude ( $A$ ) and non-dimensional flapping frequency ( $f$ ) with stiffness of the flag. [17]

By exposing the inverted flag configuration to the wing wake which is highly unsteady, the changes in frequency and amplitude of the oscillations can be quantified in a pseudo three-dimensional unsteady flow field. The piezo embedded PVDF in the inverted flag configuration was placed in the wake of SD7003 and NACA0012 airfoil to determine its performance and sensitivity with different angle of attack and downstream distance. If the dynamics of PVDF is sensitive to angle of attack and downstream distance, it will provide evidence that it can be used as an energy harvester and as a sensor for detecting changes in the wake simultaneously.

## II. Experimental Setup

### A. Wind Tunnel

All experiments mentioned in this paper were conducted at the University of Dayton Low Speed Wind Tunnel (UD-LSWT). The UD-LSWT has a 16:1 contraction ratio, 6 anti-turbulence screens and 4 interchangeable 76.2cm x 76.2cm x 243.8cm (30" x 30" x 96") test sections. The test section is convertible from a closed jet configuration to an

open jet configuration with the freestream range of 6.7m/s (20 ft/s) to 40m/s (140 ft/s) at a freestream turbulence intensity below 0.1% measured by hot-wire anemometer. The tunnel also has the ability to vary the freestream velocity profile at up to 5 Hz and over 50% velocity amplitude using a downstream shuttering system.

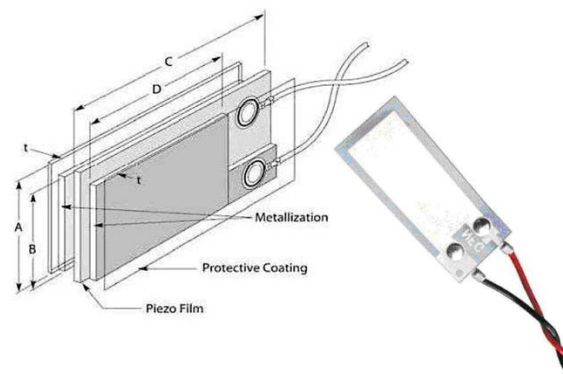
All the experiments mentioned in the paper were done in the open jet configuration where an inlet of 76.2 cm x 76.2 cm opens to a pressure sealed plenum. The effective length of the test section in the open jet configuration is 182cm (72"). A 137cm x 137cm (44" x 44") collector collects the expanded air on its return to the diffuser. A photo of the UD-LSWT open jet configuration is shown in Figure 1. The velocity variation for a given RPM of the wind tunnel fan is found using a Pitot tube connected to an Omega differential pressure transducer (Range: 0 – 6.9 kPa).

## B. Wing Models

A NACA 0012 wall-to-wall model with a chord length of 10.16 cm (4") was designed in SolidWorks and 3D printed Stratasys uPrint SE Plus printer at the University of Dayton. The 3D printed wings were carefully sanded using a 300-grit sand paper to reduce the surface roughness. Similar method was used to design and print a SD7003 wall-to-wall model with the chord length of 10.16 cm (4"). Interfaces were designed to allow the wing to change the angle of attack using Griffin motor rotary stage.

## C. Piezo embedded PVDF Material

Off-the-shelf piezo embedded PVDFs manufactured by Measurement Specialties was acquired from Mouser electronics [22]. These smart materials are intended to be used as vibration sensors to detect vibrations in a structure to which it is embedded. In this research, the sensor was used in an inverted flag configuration for extracting energy from the wake. The piezo embedded PVDF and its specifications are shown in Figure 7. For the NACA 0012 case, a piezo-embedded PVDF material with a film length of 42 mm and a thickness of 0.1 mm was used. Whereas for SD7003 case, a thin PVDF film (64  $\mu$ m) was glued using a thin film of Epoxy resin on to a 0.3 mm polymer material (Polyethylene terephthalate (PET) polymer) and was placed in the wake. The intention of bonding piezo embedded PVDF to a polymer is to perform sensitivity study on inverted flag configuration as a function of membrane length and shape. However, such sensitivity will not be discussed in this paper but will be discussed in the follow up paper. It should be noted that even though the PVDF material can generate higher voltage (10 mV to 100 V) upon actuation as shown in Figure 7, the amperage from the sensor is in the order of mA which translates to very low power output. However, with multiple PVDFs in the wake, the power output could be amplified and stored in batteries.



|                      | Units      | NACA 0012<br>Case | SD 7003<br>Case | Minimum Impedance: 1 $M\Omega$                                          |
|----------------------|------------|-------------------|-----------------|-------------------------------------------------------------------------|
| A (Film Width)       | (mm)       | 16                | 16              | Preferred Impedance: 10 $M\Omega$ and higher                            |
| B (Electrode Width)  | (mm)       | 12                | 12              | Output Voltage: 10 mV to 100 V depending on Force and Circuit Impedance |
| C (Film Length)      | (mm)       | 42                | 42              | Operating Temperature: 0° C to +70° C [32°F to 160° F]                  |
| D (Electrode Length) | (mm)       | 30                | 30              |                                                                         |
| t (Total Thickness)  | ( $\mu$ m) | 150               | 64              |                                                                         |

Figure 7 Specifications of the piezo embedded PVDF material used in NACA 0012 and SD7003 cases.



## D. Test Setup

A schematic of the test setup for NACA 0012 case is shown in Figure 8. The piezo embedded PVDF was placed at  $0.25c$ ,  $0.5c$  and  $0.75c$  downstream locations from the trailing edge of the model. The PVDF was then connected to an Agilent technologies MSO-X 3012a oscilloscope. The frequency and amplitude data was acquired at a sampling rate of 1000 Hz. The freestream velocity that rendered full amplitude oscillations were found to be 14.8 m/s.

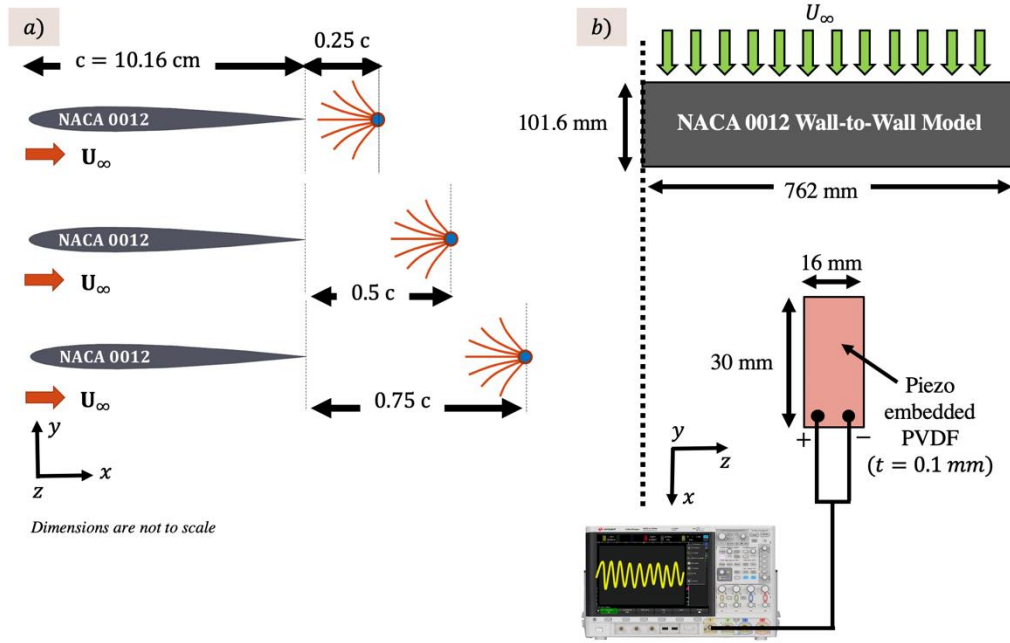


Figure 8 Schematic of a) Side view b) Top view of NACA 0012 wall-to-wall model test setup.

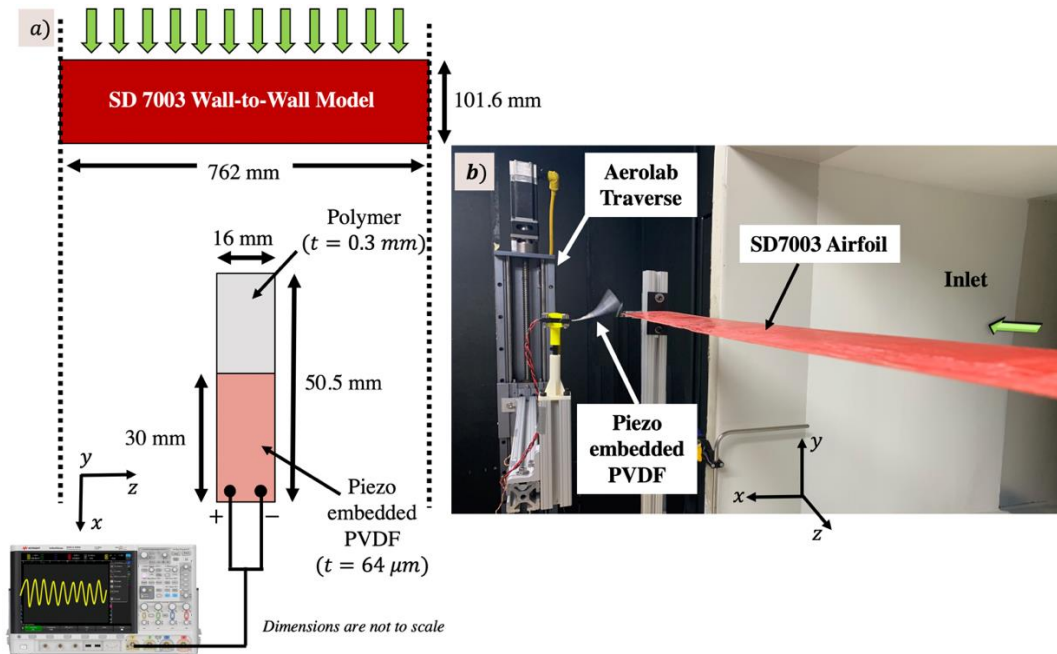


Figure 9 a) Schematic of the SD7003 wall-to-wall model test setup b) Picture of the test setup at UD-LSWT.

Similar to NACA 0012, a piezo embedded PVDF-polymer combination was tested in the wake of SD7003 wall-to-wall model. It should be noted once again that in this case, a  $64\mu\text{m}$  thick piezo embedded PVDF was attached to a thicker (0.3 mm) Polyethylene terephthalate (PET) polymer. The specifications and schematic of the PVDF-

polymer setup is shown in Figure 9a along with the picture of the test setup in Figure 9b. The freestream velocity that rendered full amplitude oscillations of the PVDF-polymer combination was determined at 11.5 m/s. The comparison between the calculated velocity from non-dimensional stiffness and the velocity at which the full amplitude oscillations was achieved will be discussed in the results section.

In this case, the PVDF-polymer combination was placed at multiple measurement locations downstream of the trailing edge of the airfoil to determine the changes in amplitude and frequency due to spatial changes in the wing wake. Two axes Aerolab traverse was used to move the PVDF-polymer combination in the X and Y axes downstream of the trailing edge of the airfoil. As shown in Figure 10, the sensor oscillations and frequency was gathered at 56 measurement locations downstream of the wing. The PVDF was moved in the y-direction in increment of  $0.125 x/c$  on both sides of the chord line making the vertical traverse length of  $0.75c$ . The PVDF was traversed along the x-direction in increment of  $0.25 x/c$  till maximum downstream distance of 2 chord lengths. Amplitude and frequency of the PVDF-polymer oscillations were obtained using Agilent technologies MSO-X 3012a oscilloscope at a sampling rate of 1000 Hz. The amplitude of the oscillations was characterized by voltage output and not the physical amplitude of the sensor.

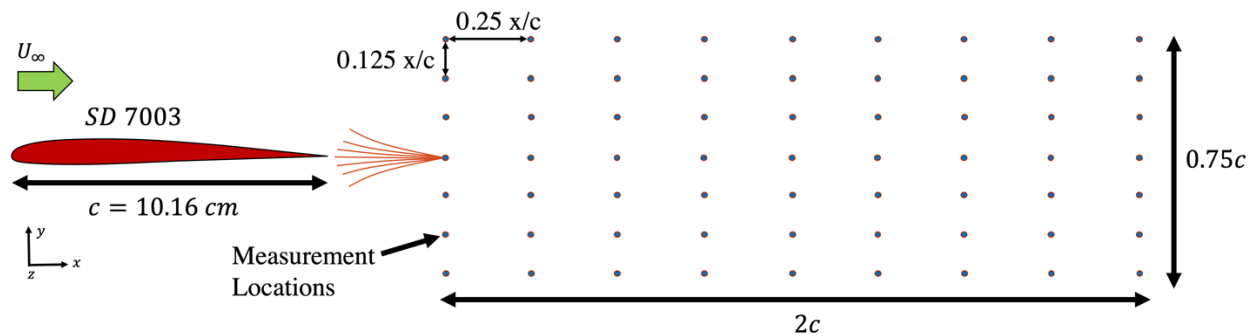


Figure 10 Downstream measurement grid locations in the wake of SD7003 wall-to-wall model.

### E. Particle Image Velocimetry

While the above mentioned experimental setup will help get an insight into the differences in the amplitude and frequency of the oscillations, in order to quantify if the presence of the PVDF affected the signature of the wake upstream, Particle Image Velocimetry was done in the wake of NACA 0012 with and without the presence of the PVDF. The results from PIV will also give insight into the relationship between the wake properties and the changes in the amplitude and frequency of the oscillations. The schematic of the PIV test setup is shown in Figure 11.

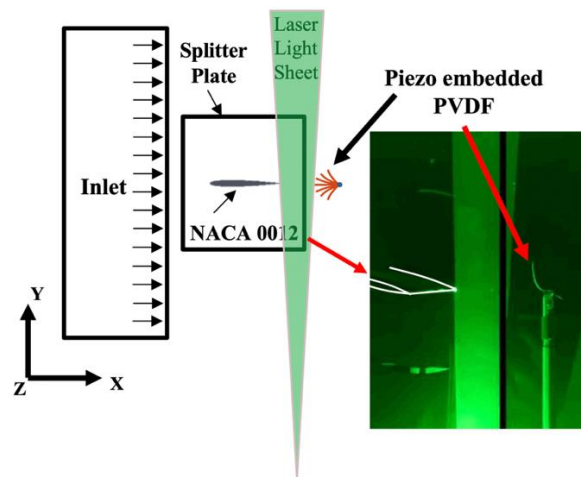


Figure 11 Schematic and picture of the PIV test setup at UD-LSWT in the wake of NACA 0012 with and without piezo embedded PVDF.

The PIV measurements were obtained using a Vicount smoke seeder with glycerin oil and a 200 mJ/pulse Nd:YAG frequency doubled laser (Quantel Twins CFR 300). A Cooke Corporation PCO 1600 camera (1600 x 1200 pixel array) with a 105 mm Nikon lens was used to capture the images. One plano-convex lens and one plano-concave



lens were used in series to convert the laser beam into a sheet. The laser and the camera were triggered simultaneously through a Quantum composer pulse generator. In each test case, over 1000 image pairs were obtained and processed using ISSI Digital Particle Image Velocimetry (DPIV) software. A total of 2 iterations were performed during PIV processing with 64-pixel interrogation windows in the first iteration and 32-pixel interrogation windows in the second iteration.

## F. Tensile Testing

In the case of SD 7003 airfoil, since the piezo embedded PVDF was bonded with the Polyethylene terephthalate (PET) polymer, tensile testing was done in accordance with ASTM D638 on the polymer. Instron Tensile Testing machine (Load Frame number: 3365) with 2kN load cell was used to determine the modulus of elasticity since it plays a vital role in determining the non-dimensional bending stiffness as shown in Equation 1. The stress-strain graph of the polymer was determined along with the modulus of elasticity at 2% strain. Five different samples of the polymer were tested. The resultant modulus of elasticity of the polymer was determined to be 2.12 GPa. It should also be noted that the modulus of elasticity of the PVDF used in the NACA 0012 case is 2 GPa as well. [23]

## IV. Results

### A. Non-dimensional Bending Stiffness

Using the dimensions of the piezo embedded PVDF tested in the wake of the airfoils and the modulus of elasticity determined from tensile testing, the non-dimensional bending stiffness was ( $\beta$ ) quantified for both the PVDFs as a function of freestream velocity. The calculated  $\beta$  values for the PVDFs used in NACA 0012 and SD 7003 cases are shown in Figure 12 as a function of freestream velocity. As the velocity increases, the non-dimensional bending stiffness decreases which results in different dynamic modes of the inverted flag configuration as shown in Figure 5. Since the PVDF used in NACA 0012 case had a shorter length, the same  $\beta$  values were achieved at higher freestream velocities when compared to the PVDF used in SD7003 case. But both PVDFs have similar trends as can be seen in the figure. Theoretically, the threshold  $\beta$  value of 0.35 where the inverted flag configuration experiences large amplitude flapping mode was predicted at a freestream velocity of 10.8 m/s for the SD7003 case using Equation 1. The same condition was predicted at a velocity of 15.4 m/s for the NACA 0012 case.

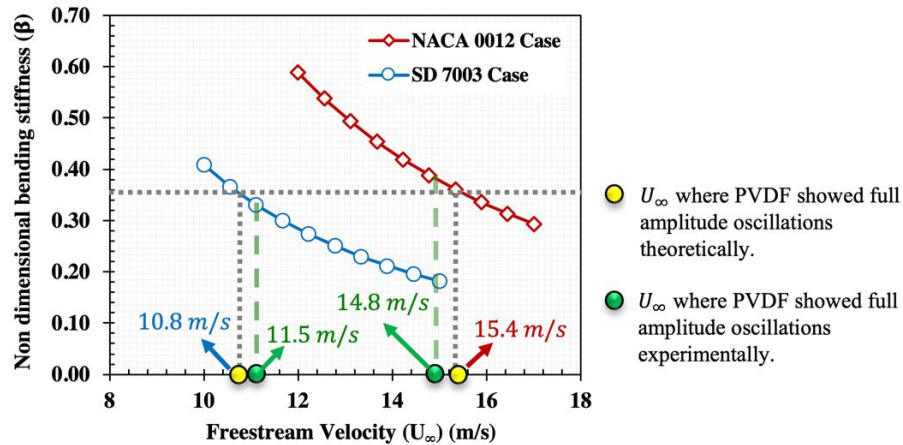


Figure 12 Variation of non-dimensional bending stiffness as a function of freestream velocity.

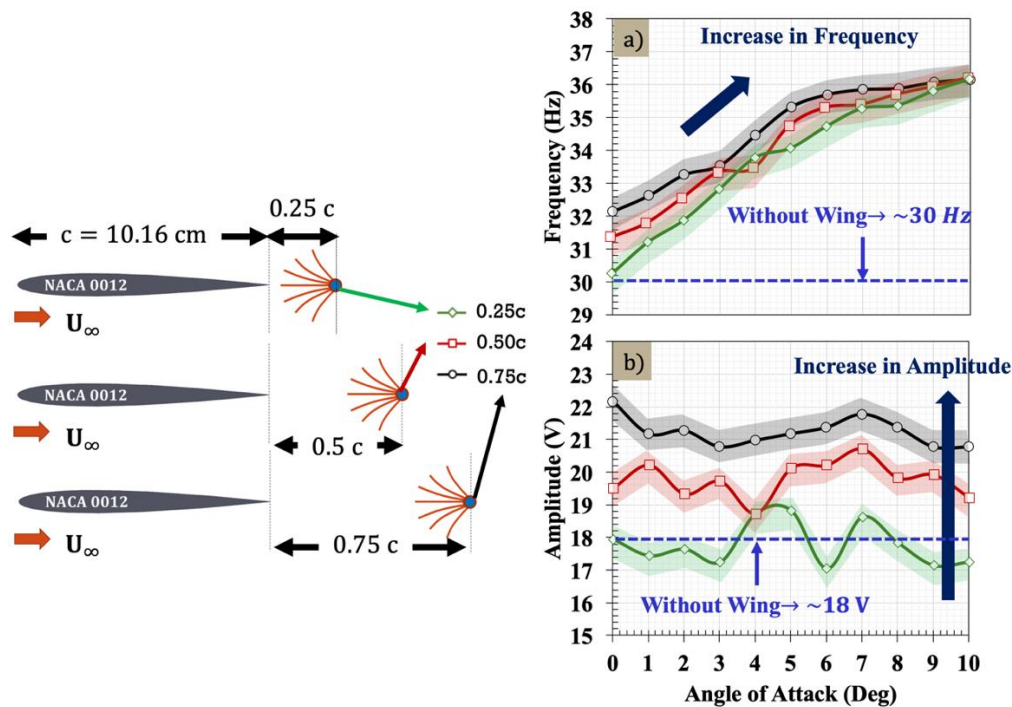
Experimentally, however, the large amplitude flapping mode for the NACA 0012 case was observed at a freestream velocity of 14.8 m/s and for the SD 7003 case, the large amplitude flapping mode were observed at a freestream velocity of 11.5 m/s. Since the wings were also placed in the same respective freestream velocity, the corresponding chord based Reynolds numbers are 100,000 for NACA 0012 airfoil and 67,000 for SD7003 airfoil. The uncertainty in the velocity measurements shown in Figure 12 is 0.1 m/s. The variation between the theoretical predictions and experimental data was 6% in the case of SD7003 and 4% in the case of NACA 0012. The comparatively larger

percentage difference in the SD7003 could be due to the attachment of the piezo embedded PVDF material with the polymer using resin.

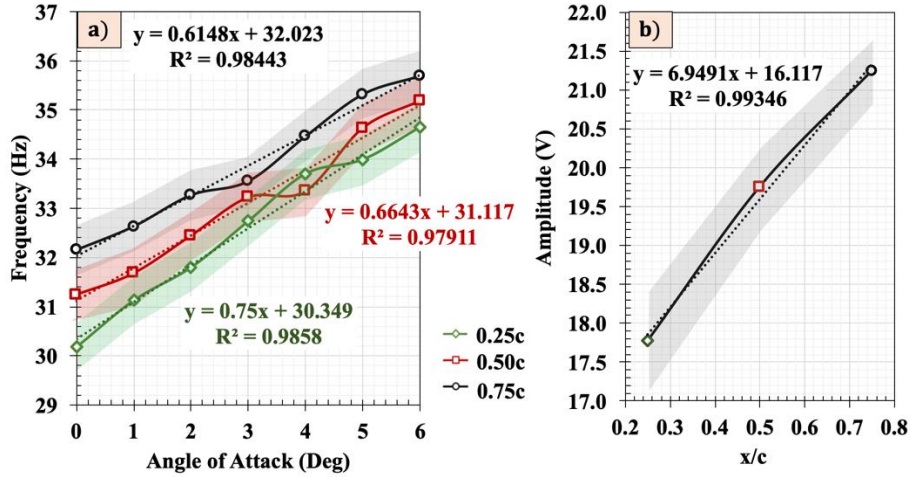
## B. PVDF Oscillations in NACA 0012 Wake:

The oscillations of the PVDF material in the wake of the NACA 0012 wall-to-wall model was quantified by measuring the frequency and the amplitude of the PVDF material as shown in Figure 13. Since the amplitude of the oscillations were characterized by the output voltage, non-dimensional frequency and amplitude could not be calculated. The PVDF material was placed at three different downstream locations from the trailing edge of the wing. The angle of attack was change from  $0^\circ$  to  $10^\circ$  in increments of  $1^\circ$ . When the PVDF was placed in the wind tunnel in a 14.8 m/s freestream velocity without the wing, full amplitude oscillations were observed. The frequency of the oscillation of the PVDF was around 30Hz. The peak to peak voltage amplitude was 18V as shown in Figure 13b. Once the PVDF was put in the wake of the wing, at  $0^\circ$ , the PVDF showed similar frequency and oscillation. But as the angle of attack is increased, the frequency of the oscillations increased almost linearly till  $7^\circ$  angle of attack and non-linearly after  $7^\circ$  angle of attack. At  $10^\circ$ , the frequency of the oscillations was found to be 36 Hz which is an increase of 20% in the frequency. As the angle of attack increases, the wing acts similar to bluff body with greater vortex shedding which in turn increases the frequency of the oscillations. It should be noted that the location of the PVDF was not changed with increase in angle of attack. It is also found that the as frequency increases with increase in distance downstream. This can be attributed due to increase in the wake width with increase in distance downstream.

Similar behavior is seen in the amplitude of the PVDF oscillations shown in Figure 13b. The amplitude is quantified in terms of peak to peak voltage output from a single PVDF sensor in the wake. The amplitude of the PVDF was also quantified without the wing. The oscillation amplitude of the PVDF without the presence of the wing was found to be 18V. There are small variations in the amplitude as a function of angle of attack for all three downstream distances and is not as pronounced as change in frequency. But the variation in amplitude doesn't show a constant increase or decrease with angle of attack. However, the amplitude of the oscillations increased as a function of downstream distance. Placed at a downstream distance of  $0.25c$ , the mean amplitude of PVDF is similar to the amplitude without the wing at 18 V. With the increased downstream distance of  $0.5c$ , the mean amplitude increased to 20 V which is about 11% increase from the  $0.25c$  downstream case. With the increased distance of  $0.75c$ , the mean amplitude further increased to 20 V which is around 20% increase from the  $0.25c$  baseline case.



**Figure 13** Variation of a) frequency and b) amplitude of the PVDF oscillations with angle of attack at different downstream locations. The frequency increases with increase in angle of attack but the amplitude remains constant. But both the frequency and the amplitude increases with increase in downstream distances.



**Figure 14 a) Variation of PVDF oscillation frequency with angle of attack b) Variation of amplitude of PVDF oscillation as a function of downstream distance.**

Figure 14 shows a direct correlation between the angle of attack and oscillation frequency and amplitude. The linear increase in the frequency till 7° angle of attack is represented clearly in Figure 13a which surprisingly resembles the linear increase in coefficient of lift with angle of attack. The slope of the frequency changes with angle of attack determined (by the linear regression equation) decreases with increase in downstream distance. A linear relationship is also found in the variation of amplitude of the oscillations with the downstream distance as shown in Figure 14b. Despite the highly complex flow structures that arises from the wake-PVDF interactions, the linear relationship between the frequency and angle of attack is surprising. This linear dependency could arise from the coherent structures present in the wake of the wing. In order to examine the relationship further, changes in coherent structures in the wake was correlated with the frequency increase in the section below.

### C. Relationship between Frequency and Coherent Structures in NACA 0012 Wake

As seen in the literature review sections, the conventional flexible trailing edge extensions ([8], [9]) are excited by the vortex shedding from the wake and that the frequency of the oscillations of the membrane correspond with the vortex shedding frequency. In order to understand the role of vortex shedding on the inverted flag configurations, two-point correlation of  $u'$  and  $v'$  velocity components was determined in the wake of the wing with and without the presence of the piezo embedded PVDF. The two-point correlation allows to determine the length scales associated with the coherent turbulent motions (vortex shedding) in the wake which are responsible for pressure fluctuations, sound generation and structural vibration. Bendat and Piersol [23] defined the two-point correlation as

$$\rho_{u_i u_j}(X_1, X_2, \tau) = \frac{[u'_i(X_1, t) * u'_j(X_2, t + \tau)]}{\sqrt{u'_i(X_1)^2} \sqrt{u'_j(X_2)^2}} \quad (3)$$

where  $X_1$  and  $X_2$  are two spatial locations in the PIV field of view,  $\tau$  is the time delay (which is chosen to be zero for the results shown in Figure 15),  $u'$  represents the fluctuating velocities in  $i$  and  $j$  direction. The reference point  $X_1$  was chosen to be at the mid-point of the upper surface shear layer at the trailing edge of the wing.

The two-point correlation contours of fluctuating velocity in the streamwise direction ( $\rho_{uu}$ ) at angles of attack of 0°, 2°, 6° and 8° is shown in Figure 15. It can be seen that as the angle of attack increases, the coherent structures in the wake of the wing increases which is observed by the alternating red and blue patterns in the contour. There are 7 coherent structures in the upper surface wake at 0° angle of attack whereas with increase in angle of attack, the number of coherent structures increased to 10 at 8° angle of attack. These patterns indicate the presence of vortices in the wake which are shed from the upper and lower surface of the wing. Surprisingly, the changes in the wake signature was picked up the inverted PVDF as an increase in frequency which implies that the piezo embedded PVDF in the inverted flag configuration can be used for wake sensing purposes. In order to determine if the presence of inverted PVDF changed the upstream wake signature, comparisons were made using wake signatures. The momentum deficit and two-point correlations were compared between the cases with and without the presence of PVDF. The results are shown in the section below.

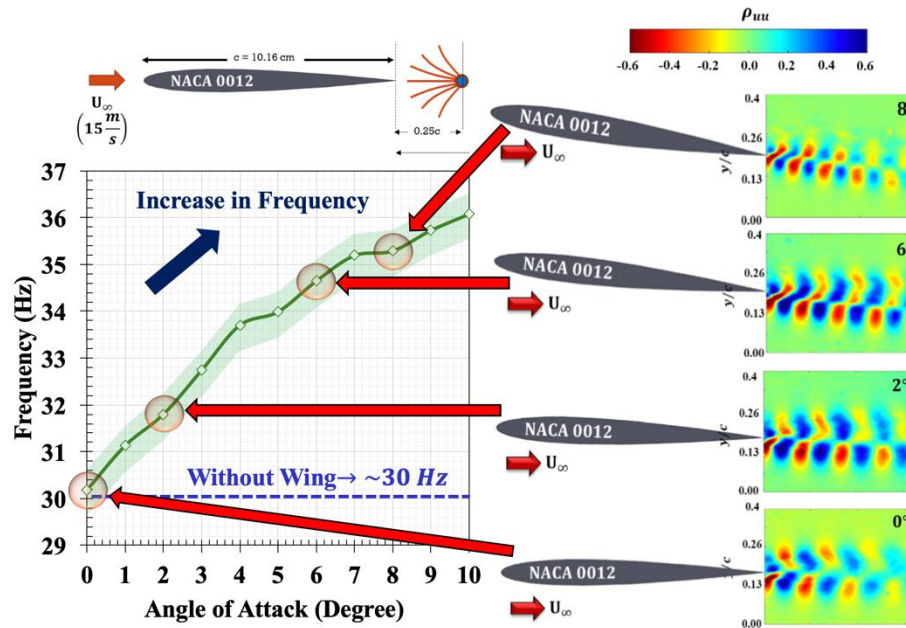


Figure 15 Variation of PVDF oscillation frequency with angle of attack due to increase in coherent structures in the wake of NACA 0012 airfoil.

## D. Influence of PVDF on the Wake Structure

### 1. Momentum Deficit

The increase in the PVDF frequency with the increase in coherent length scales shown in Figure 15 is only valid if the presence of PVDF did not alter the upstream wake structure itself.

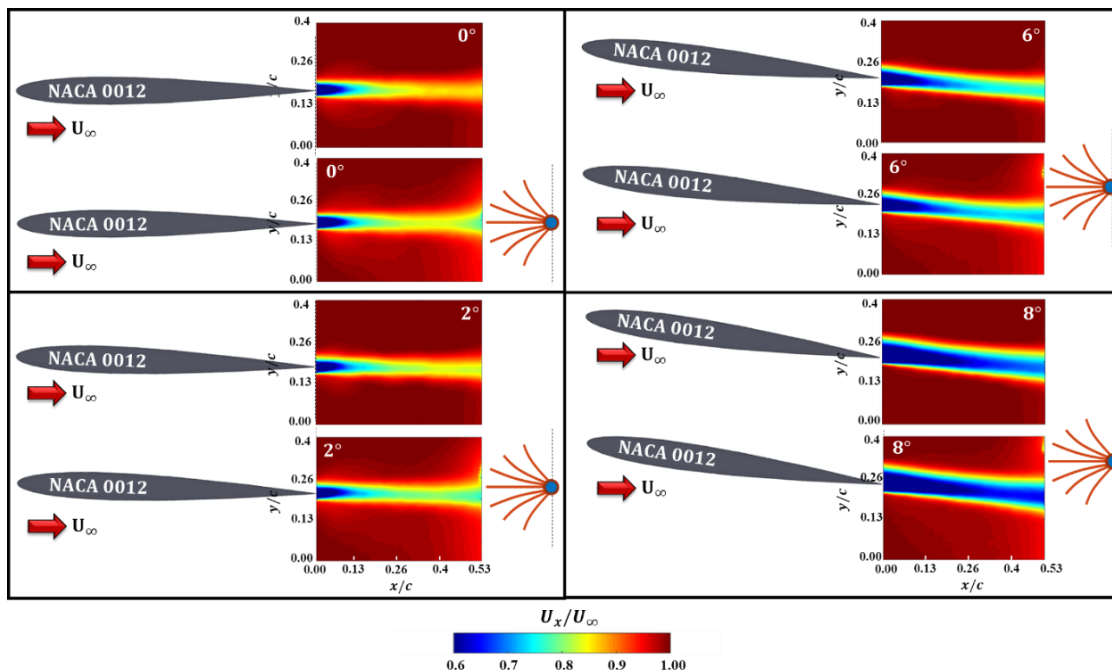
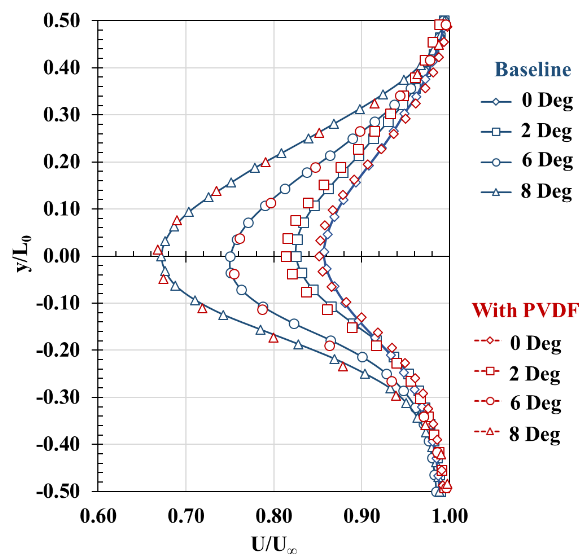


Figure 16 Momentum deficit contours in the wake of NACA 0012 airfoil with and without the inverted PVDF.

Therefore, the momentum deficit (Figure 16) and two-point correlation contours (Figure 18) are compared in the NACA 0012 wake with and without the piezo embedded PVDF. From Figure 16, at 0° and 2° angles of attack, the momentum deficit near the leading edge of the PVDF increases by 2% when compared to the baseline. As the

angle of attack is increased, the momentum deficit behind the trailing edge increases as expected but the momentum deficit ahead of the PVDF remains unchanged. Similar trends are seen in the  $6^\circ$  and  $8^\circ$  angle of attack as well where the momentum deficit is very similar in the wake with and without the PVDF. This could be due to the fact that the location of the PVDF was not changed with angle of attack as indicated in Figure 16. As the downwash increases with the angle of attack, only the wake from the upper surface is affected. The differences in the momentum deficit between the baseline and the PVDF cases can be seen clearly in the momentum deficit profiles shown in Figure 17.

The momentum deficit profiles shown in Figure 17 indicate small differences with and without the presence of the PVDF. These profiles were obtained from the momentum deficit contour at the center of the field of view. The increase in momentum deficit with angle of attack can be seen clearly in Figure 17 for both the cases. At lower angles of attack, the presence of PVDF increased the peak momentum deficit by 2%. At higher angle of attack, the momentum deficit profiles show similar variation between the two cases. Any changes in the momentum deficit observed at high angle of attack falls within the uncertainty of the PIV measurements.



**Figure 17 Momentum deficit profiles in the wake of NACA 0012 airfoil with and without PVDF at different angles of attack.**

## 2. Two-Point Correlations

The normalized two-point correlation functions with zero-time delay of the streamwise ( $u'$ ) and transverse ( $v'$ ) fluctuating velocities are shown in Figure 18 and Figure 19 respectively for angles of attack of  $0^\circ$ ,  $2^\circ$ ,  $6^\circ$  and  $8^\circ$ . In each case, the reference point ( $X_1$ ) in Equation 3 is chosen to be at the center of the upper shear layer. The contour images show extensive coherent structures of alternating positive and negative correlation values for both components of fluctuating velocity. Specifically, spatially alternating regions of positive and negative correlation are indicative of the spatially and temporally periodic motions which is related to the tonal character of fluctuations in the flowfield at the frequency of vortex shedding.

Very subtle differences can be observed in the correlations with and without the presence of the PVDF in the wake. The correlation of the velocity closer to the PVDF decreases slightly when compared to the baseline case at  $0^\circ$  and  $2^\circ$  angle of attack cases. At higher angles of attack, the correlations of the U and V fluctuating velocities between the baseline and the PVDF case is almost identical. This could be due to the PVDF not being in the center of the wake.

The coherent structures are highlighted clearly in the two-point correlation contours of transverse velocity components shown in Figure 19. The alternating pattern of red and blue bands indicate increase and decrease in velocity when compared to the velocity at the reference point  $X_1$  respectively. The coherent structures clearly indicate that the presence of the PVDF did not change the nature of the rollup of the coherent structures across all angles of attack. Similar number of correlation bands can be seen for all angles of attack between the wake without the presence of PVDF and with the presence of the PVDF. Since both the mean and fluctuating component of the wake shows similar variation between the two cases (with and without the PVDF), it conclusively shows that the presence of PVDF did not affect the nature of the wake hence it can be used for sensing changes in the wake signature of a wing.



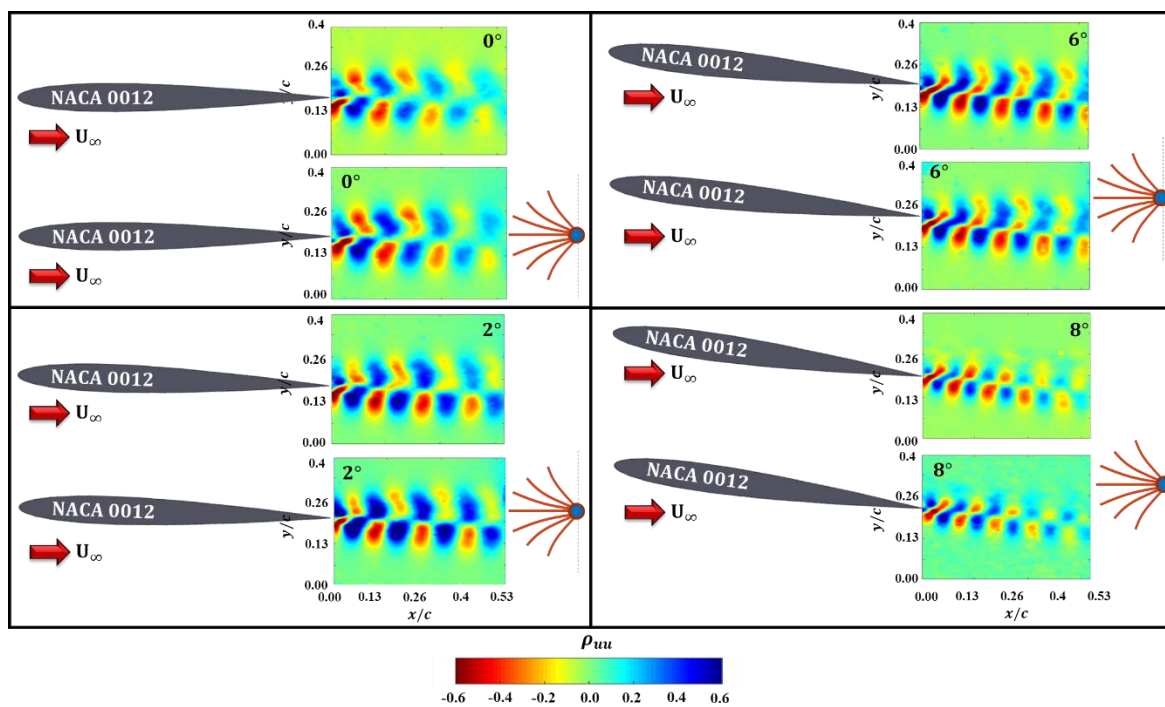


Figure 18 Two-point correlations of  $u'$  (streamwise fluctuating velocity component) in the wake of NACA 0012 with and without PVDF.

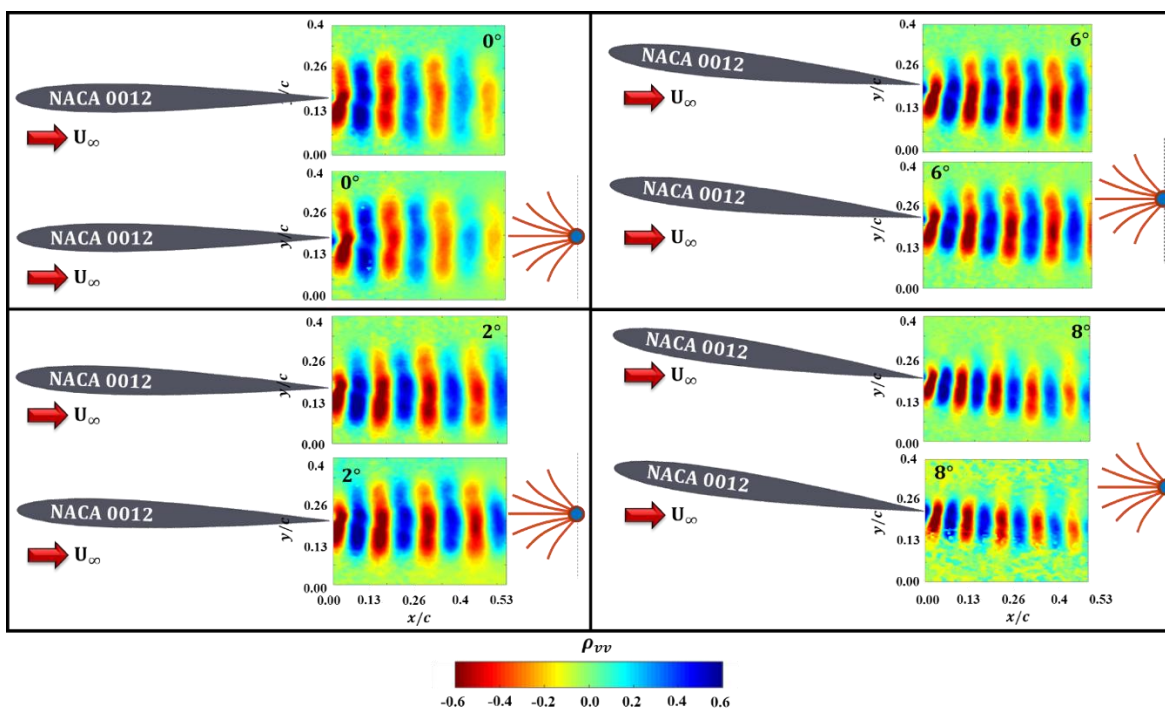


Figure 19 Two-point correlations of  $v'$  (transverse fluctuating velocity component) in the wake of NACA 0012 with and without PVDF.

### E. Performance of piezo embedded PVDF – Polymer sensor in Freestream

Before testing the PVDF - Polyethylene terephthalate (PET) polymer combination shown in Figure 9a in the wake of the SD7003 airfoil, its performance was quantified by placing it in the freestream to determine the average peak-to-peak voltage output and the oscillation frequency. The average peak-to-peak Voltage is determined by taking the peak voltages in the time history of the voltage output from the PVDF and averaging them for 2 seconds with sampling rate of 1000 Hz. The oscillation frequency was determined by taking the Fast Fourier Transform of the voltage signal similar to the frequencies shown in the previous section for the NACA 0012 case. The variation of the voltage and frequency of the inverted PVDF – Polymer configuration is shown in Figure 20. The sensor was placed in the freestream range between 10 m/s and 14.7 m/s to which the corresponding non-dimensional stiffness values were calculated to be 0.36 and 0.21 respectively. At these velocities, the PVDF-polymer configuration underwent large amplitude flapping mode at 11.5 m/s ( $\beta = 0.34$ ). With increase in freestream velocity, the large amplitude flapping reduced to large deflection deformed flapping mode at velocity of 14.7 m/s ( $\beta = 0.20$ ) as indicated in Figure 20. The frequency of the oscillations varied highly non-linearly as a function of  $\beta$ . High frequency in the PVDF-polymer oscillations was observed at  $\beta$  value of 0.26. However, the results from Figure 20 matches the results shown in Figure 6 where similar non-linear behavior was seen as a function of  $\beta$ .

It was interesting to see that the higher peak-to-peak voltage output of 18 V was seen at a lower  $\beta$  value of 0.2 when the sensor was experiencing large deflection deformed flapping mode and not large amplitude flapping model which was counterintuitive. It should be noted that the mean voltage of the large amplitude flapping mode is close to zero since the voltage variation with time was symmetrical about the horizontal axis. However, in the case of the large deflection deformed flapping mode, the higher mean voltage was observed since the PVDF-polymer deflections were confined to the positive quadrant alone.

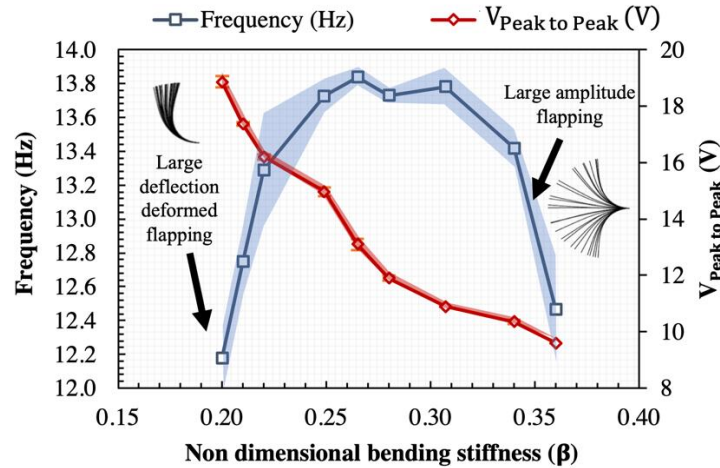


Figure 20 Variation of PVDF-Polymer frequency and peak-to-peak voltage output from PVDF-Polymer combination as a function of non-dimensional bending stiffness ( $\beta$ ).

### F. Voltage Root Mean Square (RMS) in the Wake of SD7003 Airfoil

Once the performance of the PVDF-polymer combination was quantified in the freestream, it was then placed in the wake of SD7003 wall-to-wall model to determine the changes in oscillating frequency and output voltage RMS as a function of angle of attack and downstream distance. The sensor was placed at 56 grid point locations downstream of the wing as shown in Figure 10. The Voltage RMS was calculated using Equation 4.

$$V_{RMS} = \sqrt{\frac{\sum (V_{Instant} - V_{mean})^2}{n}} \quad (4)$$

where  $V_{Instant}$  is the instantaneous voltage,  $V_{mean}$  is the mean voltage, and  $n$  is the number of data points collected. The  $V_{mean}$  for most the cases was closer to zero since the PVDF-polymer sensor underwent large amplitude flapping mode in the wake of the wing. A normalized Voltage RMS contour from the measurement locations as a function different angles of attack is shown in Figure 21. The Voltage RMS in the wake of the wing was normalized with the

Voltage RMS when the PVDF-polymer sensor was placed in the freestream. The contours show very interesting patterns which captures some of the wake characteristics. The voltage RMS contour is so surprisingly stratified that distinct red and blue regions indicate the upper and lower surface shear layers from the trailing edge of the wing. The normalized voltage RMS is higher in the upper surface shear layer wake as shown in red regions of the contour and comparatively lower in the lower surface shear layer wake. This states that for energy harvesting purposes, the sensor performs better if it is placed in the upper surface shear layer. When the PVDF was placed at the mid-wake, the normalized voltage RMS was similar to that of the freestream as indicated in white regions of the contour. The voltage RMS contours also indicates changes in downwash angle of the wake as a function of angle of attack. This changes in downwash angle in the voltage RMS contour is indicated by the dotted black line in the middle of each contours. This dotted line was drawn by closely following the white regions of the contour expect for the  $6^\circ$  angle of attack case. Due to the location of the grid measurement points and the downwash angle, it is possible that the sensor was not placed exactly at the wake center at higher angle of attack. Irrespective of the resolution of the grid location, Figure 21 shows remarkable correlation between the wake features and voltage RMS obtained from the sensor. The increase in voltage RMS with downstream distance mirrors the results seen in NACA 0012 case in Figure 13b and Figure 14b as well. These results conclusively indicate that the inverted flag configuration could deliver higher voltage output under unsteady conditions when compared to freestream conditions. It is hypothesized that the increase in voltage RMS in the upper surface could be due to the increased shear and velocity from the upper surface of the wing.

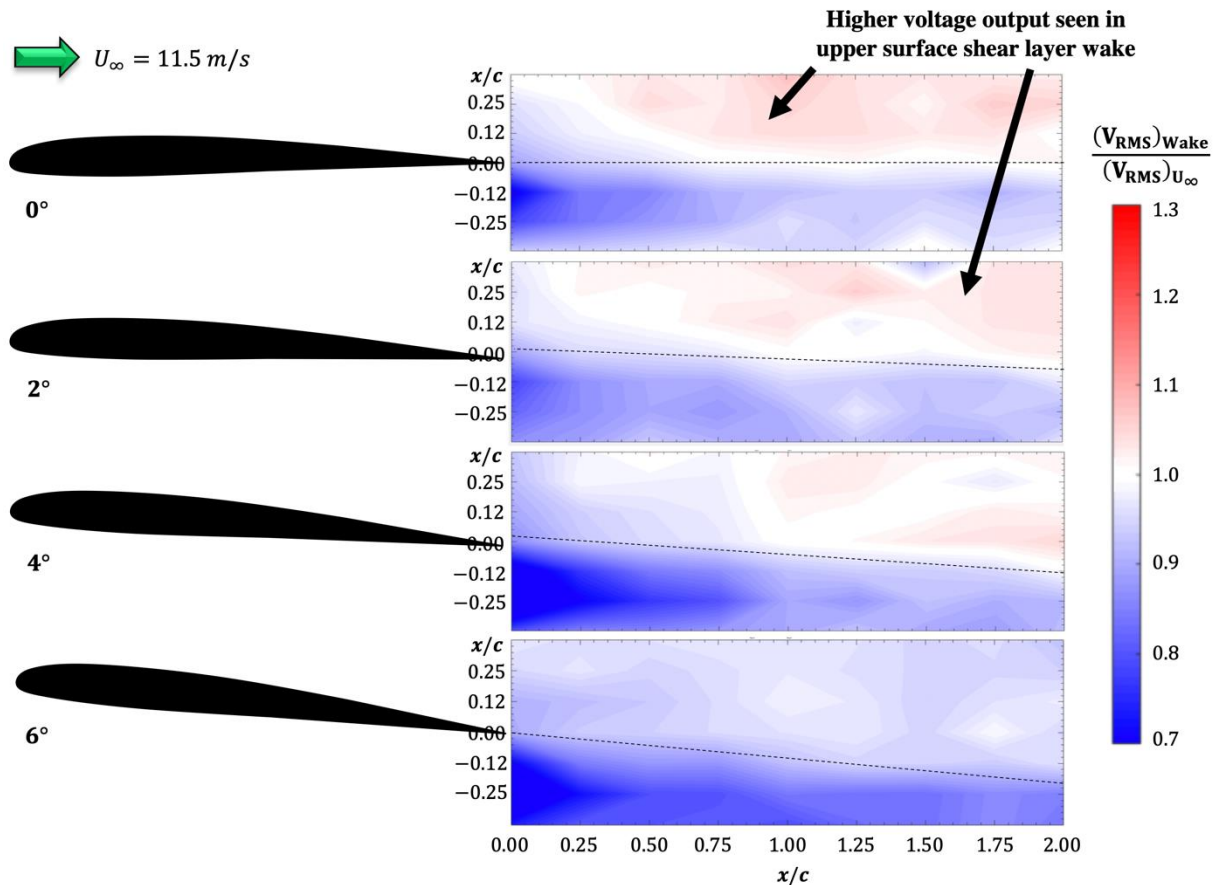


Figure 21 Normalized Voltage RMS contours in the wake of SD7003 airfoil for multiple angles of attack. A surprising pattern in the voltage output was seen where  $V_{RMS}$  increased in the upper surface shear layer and decreased in the lower surface shear layer.

## V. Conclusions

The use of inverted flexible piezo embedded PVDF for energy harvesting and wake sensing purposes were investigated at the University of Dayton Low Speed Wind Tunnel. Two different PVDFs with different thickness and lengths were tested with two different airfoils: NACA 0012 and SD7003 at Reynolds numbers of 100,000 and 60,000 respectively. These Reynolds numbers were chosen from the velocity at which the PVDF sensor achieved large amplitude flapping mode. From the experiments conducted, key conclusions are:

1. The inverted flag configuration is sensitive to unsteady flows. This dependency is shown by the increase in oscillation frequency and amplitude as a function of angle of attack and the downstream distance from the trailing edge of the airfoil. The oscillation frequency increased linearly with increase in angle of attack similar to the linear trend of lift coefficient with angle of attack. The peak to peak voltage output was increased with increase in downstream distances from the trailing edge of the wing.
2. The increase in sensor frequency with angle of attack could be due to the increase in coherent structures in the wake of the wing confirmed by the two-point correlations of fluctuating  $u'$  and  $v'$  velocity components.
3. Even with high unsteady wake, the presence of piezo embedded PVDF did not affect the upstream wake signature as evidenced by the momentum deficit contours and profiles as well as the two-point correlations of the fluctuating velocity components.
4. Higher peak-to-peak voltage output was seen when the PVDF-polymer configuration experienced large deflection deformed flapping mode at non-dimensional bending stiffness value of 0.21.
5. It is highly surprising that the voltage RMS contour in the wake of SD7003 airfoil shows distinct wake characteristics such as upper and lower surface shear layers and downwash angle. Higher voltage RMS values were observed in the upper surface shear layer wake when compared to the lower surface shear layer wake.

The increase in voltage output as well as the sensitivity of the sensor to the wake location and angle of attack changes in both NACA 0012 and SD7003 case shows that the inverted flag configuration can be used as an energy harvester as well as a sensor to detect changes in the wake signature.

## VI. Future Work

Even though the results indicate the piezo embedded PVDF has the potential to be used as an energy harvester and as a sensor to detect changes in the wake, the net power output of a single PVDF is very low due to lower amperage production. In order to determine the net power output in the Wattage, a voltage divider circuit must be designed and connected to the PVDF. Apart from measuring direct power output, multiple PVDF-polymer type configurations will be connected in series or in parallel and placed in the wing wake. Efforts are already underway to quantify the power output from two inverted PVDF sensors placed side-by-side in the wake of the wing. In future, sensitivity studies will also be done in the geometry and planform (rounded leading edge, elliptical planform, tapered leading edge, etc.) of the inverted PVDF configurations.

## VII. Acknowledgements

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